

Distinct-symbol bilinear commutators: an affirmative answer for the sum and a counterexample for the iterate

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11 August 2026

Abstract

We resolve the two final questions of arXiv:1707.01639 in the standard real-symbol setting of the paper. For a homogeneous, nondegenerate bilinear convolution Calderón–Zygmund operator, boundedness of $[b_1, T]_1 + [b_2, T]_2$ forces $b_1, b_2 \in \text{BMO}$. The proof first detects $b_1 + b_2$, then subtracts its ordinary commutator and detects b_1 from the difference of the two slots. In contrast, the iterated-commutator question is false as printed: a constant inner symbol annihilates the entire iterate, while the other symbol may be outside BMO.

1 Setting and result

Let

$$T(f, g)(x) = \iint_{\mathbb{R}^n \times \mathbb{R}^n} K(x - y_1, x - y_2) f(y_1) g(y_2) dy_1 dy_2$$

off the common support. We use the kernel hypotheses intended in the source: K is a continuous bilinear Calderón–Zygmund kernel, homogeneous of degree $-2n$, and nonzero away from the origin in $\mathbb{R}^{2n}$. The last condition follows in the source from the assumption that $1/K$ has an absolutely convergent Fourier series on every cube avoiding the origin. We also use the standard sufficiency theorem, proved in the source, that $a \in \text{BMO}$ implies boundedness of each $[a, T]_j$ in the full bilinear range.

For locally integrable symbols set

$$\mathcal{S}_{b_1, b_2} = [b_1, T]_1 + [b_2, T]_2, \quad \mathcal{P}_{b_1, b_2} = [b_2, [b_1, T]_1]_2.$$

Theorem 1 (Split resolution of Problems A and B). *Let b_1, b_2 be real-valued and locally integrable, let $1 < p_1, p_2 < \infty$, and put $1/q = 1/p_1 + 1/p_2$ (so $q > 1/2$).*

1. *If $\mathcal{S}_{b_1, b_2} : L^{p_1} \times L^{p_2} \rightarrow L^q$ is bounded, then $b_1, b_2 \in \text{BMO}$.*
2. *Boundedness of $\mathcal{P}_{b_1, b_2}$ does not imply $(b_1, b_2) \in \text{BMO} \times \text{BMO}$, even if the two symbols are distinct. Indeed, one may take $b_1 = 0$ and $b_2(x) = x_1$.*

The first assertion also holds for complex symbols whenever the kernel is real-valued up to a fixed scalar phase. For arbitrary complex kernels, the same proof is complete for real symbols; the source’s questions and examples are normally read in this real-symbol setting.

2 A separated-cube sector lemma

Lemma 1 (Uniform sector). *There are a vector $z \in \mathbb{R}^n$, a fixed number $R > 1$, a unimodular σ , and $c_K > 0$ with the following property. If P is a cube of side length ℓ , and $Q = P + R\ell z$, then*

$$\operatorname{Re}(\sigma K(x - y_1, x - y_2)) \geq c_K \ell^{-2n} \quad (x \in Q, y_1, y_2 \in P).$$

The roles of a prescribed cube and its translate may be reversed.

Proof. Choose $z \neq 0$. Nonvanishing gives $K(z, z) \neq 0$; rotate it by a unimodular σ so that $\sigma K(z, z) > 0$. If $x \in Q$ and $y_i \in P$, then

$$\ell^{2n} K(x - y_1, x - y_2) = K(Rz + u_1, Rz + u_2) = R^{-2n} K(z + u_1/R, z + u_2/R),$$

where $|u_i| \leq C_n$. Continuity at (z, z) makes the last expression stay in a fixed open half-plane when R is sufficiently large. Since R is then fixed, the asserted lower bound follows. Translating in the opposite direction gives the reversed geometry. $\square$

3 Problem A: recovery of both symbols

Write $s = b_1 + b_2$, and let A denote the operator norm of $\mathcal{S}_{b_1, b_2}$.

Step 1: recover the sum

Fix an arbitrary cube Q of side ℓ , and use Lemma 1 to choose a separated cube P of the same size on which the two inputs will be supported. Let m_i be a median of b_i on P , and define

$$E_i^- = \{y \in P : b_i(y) \leq m_i\}, \quad E_i^+ = \{y \in P : b_i(y) \geq m_i\}.$$

Both sets have measure at least $|P|/2$. Put $m = m_1 + m_2$. The kernel formula for the summed commutator is

$$\mathcal{S}_{b_1, b_2}(f, g)(x) = \iint K(x - y_1, x - y_2) (s(x) - b_1(y_1) - b_2(y_2)) f(y_1) g(y_2) dy_1 dy_2.$$

For $x \in Q \cap \{s \geq m\}$ and $y_i \in E_i^-$, the parenthesis is a nonnegative real number at least $s(x) - m$. Lemma 1 therefore gives

$$(s(x) - m)_+ \leq \frac{C\ell^{2n}}{|E_1^-||E_2^-|} \left| \mathcal{S}_{b_1, b_2}(\mathbf{1}_{E_1^-}, \mathbf{1}_{E_2^-})(x) \right|.$$

Taking the $L^q(Q)$ quasi-norm and using boundedness yields

$$\|(s - m)_+\|_{L^q(Q)} \leq CA\ell^{2n} |E_1^-|^{1/p_1-1} |E_2^-|^{1/p_2-1} \leq CA|Q|^{1/q}.$$

The upper median sets give the same estimate for $(s - m)_-$. Hence

$$\sup_Q \inf_{c \in \mathbb{R}} \left(\frac{1}{|Q|} \int_Q |s - c|^q \right)^{1/q} \leq CA.$$

The L^q -oscillation characterization of BMO is valid for every $q > 0$ (for $q < 1$, this is precisely the $\operatorname{BMO}_q = \operatorname{BMO}$ lemma proved in the target paper). Thus $s \in \operatorname{BMO}$.

Step 2: recover one symbol from the slot difference

Since $s \in \text{BMO}$, the ordinary commutator $[s, T]_2$ is bounded in the same range. Consequently

$$\mathcal{D}_{b_1} := \mathcal{S}_{b_1, b_2} - [s, T]_2 = [b_1, T]_1 - [b_1, T]_2$$

is bounded. Its kernel identity has the crucial cancellation

$$\mathcal{D}_{b_1}(f, g)(x) = \iint K(x - y_1, x - y_2)(b_1(y_2) - b_1(y_1))f(y_1)g(y_2) dy_1 dy_2. \quad (1)$$

Now fix an arbitrary cube P , let m be a median of b_1 there, and set

$$E = \{b_1 \leq m\} \cap P, \quad F = \{b_1 \geq m\} \cap P.$$

Again $|E|, |F| \geq |P|/2$. Choose the separated output cube Q from Lemma 1. On $E \times F$, the factor in (1) is nonnegative, so for every $x \in Q$,

$$|\mathcal{D}_{b_1}(\mathbf{1}_E, \mathbf{1}_F)(x)| \geq c_K \ell^{-2n} I_P, \quad I_P := \int_E \int_F (b_1(y_2) - b_1(y_1)) dy_2 dy_1.$$

The operator bound, $|Q| = |P|$, and $1/q = 1/p_1 + 1/p_2$ imply

$$c_K \ell^{-2n} I_P |Q|^{1/q} \leq C |E|^{1/p_1} |F|^{1/p_2} \leq C |P|^{1/q},$$

and hence $I_P \leq C |P|^2$. On the other hand,

$$I_P = |E| \int_F (b_1 - m) + |F| \int_E (m - b_1) \geq \frac{|P|}{2} \int_P |b_1 - m|.$$

Therefore

$$\frac{1}{|P|} \int_P |b_1 - m| \leq C$$

uniformly in P , proving $b_1 \in \text{BMO}$. Finally $b_2 = s - b_1 \in \text{BMO}$. This proves Theorem 1(1).

4 Problem B: a universal counterexample

Take

$$b_1(x) = 0, \quad b_2(x) = x_1.$$

The symbols are distinct, $b_1 \in \text{BMO}$, and $b_2 \notin \text{BMO}$, since its mean oscillation on cubes of side L grows proportionally to L . Nevertheless

$$[b_1, T]_1 = 0, \quad \mathcal{P}_{b_1, b_2} = [b_2, [b_1, T]_1]_2 = [b_2, 0]_2 = 0.$$

The zero bilinear operator is bounded between every pair of quasi-Banach function spaces. Thus Problem B has a negative answer under every kernel and exponent hypothesis in the source. The obstruction is intrinsic: an iterated commutator only sees the joint oscillation of the tuple and cannot force each factor into BMO without an individual nondegeneracy assumption.

5 Scope and checks

The affirmative argument uses exactly two structural inputs beyond boundedness of T : a fixed-phase separated-cube lower bound, supplied by homogeneity and nonvanishing, and the already-known BMO sufficiency for ordinary commutators. It does not use the Fourier-series representation of $1/K$.

For complex-valued symbols and a real kernel (or a kernel with one fixed global phase), apply the real argument to real and imaginary parts, using complex conjugation and linearity. A variable-phase complex kernel with complex symbols requires a finite-sector version of the median lemma; that corner is not claimed here. Problem B's counterexample has no such restriction.

References

- [1] D. Wang, J. Zhou, and Z. Teng, *Characterizations of weighted BMO space and its application*, arXiv:1707.01639.